

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4

Total Marks:40

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round

- Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.
5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Experiment 1

Aim

To design a simple LAN using two PCs connected through a switch, verify connectivity using the **ping** command, and analyze the Ethernet frame using Wireshark.

Procedure

Two PCs were connected to a switch in Cisco Packet Tracer. The IP addresses were configured as:

- **PC1:** 192.168.1.1 /24
- **PC2:** 192.168.1.2 /24

After assigning the IP addresses, the connectivity was verified using the **ping** command. Wireshark was used to capture the packets exchanged during communication.

Observation

The ping request and reply were successful, indicating that both PCs were communicating properly. In the captured Ethernet frame, the following fields were observed:

- **Source MAC Address:** MAC address of PC1
- **Destination MAC Address:** MAC address of PC2
- **EtherType:** IPv4 (0x0800)
- **Frame Check Sequence (FCS):** Used to detect transmission errors

Result and Analysis

The Data Link Layer uses MAC addresses to identify devices within the same local network. Before sending data, the sender includes both the source and destination MAC addresses in the Ethernet frame. The FCS field helps detect transmission errors, ensuring that corrupted frames are discarded. Since the MAC addresses were correctly identified, the frame reached the destination successfully and the ping command completed without errors.

Experiment 2

Aim

To create two LANs connected through a router, configure IPv4 addressing, and analyze the IPv4 header fields using Wireshark.

Procedure

Two separate LANs were created in Cisco Packet Tracer and connected using a router. The following IP configuration was used:

LAN 1

- PC1: 192.168.1.2/24
- Gateway: 192.168.1.1

LAN 2

- PC2: 192.168.2.2/24
- Gateway: 192.168.2.1

The router interfaces were configured with the gateway addresses. After configuration, the ping command was executed from PC1 to PC2, and the packets were captured using Wireshark.

Observation

The ping was successful, showing that communication between the two networks was established. The IPv4 packet contained the following fields:

- **Source IP Address:** 192.168.1.2
- **Destination IP Address:** 192.168.2.2
- **Time-To-Live (TTL):** Decreased by one after passing through the router
- **Protocol Number:** 1 (ICMP)
- **Header Length:** 20 bytes

Result and Analysis

The Network Layer is responsible for routing packets between different networks. It uses IP addresses to identify the sender and receiver. The router examines the destination IP address and forwards the packet to the correct network. The TTL value decreases whenever the packet passes through a router, preventing packets from circulating endlessly in the network.

Experiment 3

Aim

To study the Address Resolution Protocol (ARP) by clearing the ARP cache, sending a ping request, and analyzing ARP packets using Wireshark.

Procedure

A LAN with multiple hosts connected through a switch was created in Cisco Packet Tracer. The ARP cache on one host was cleared before sending a ping request to another host. Wireshark was used to capture the ARP Request and ARP Reply packets.

Observation

The following packets were captured:

ARP Request

- Sender IP Address: 192.168.1.2
- Sender MAC Address: MAC address of PC1
- Target IP Address: 192.168.1.3
- Target MAC Address: Unknown (00:00:00:00:00:00)

ARP Reply

- Sender IP Address: 192.168.1.3
- Sender MAC Address: MAC address of PC2
- Target IP Address: 192.168.1.2
- Target MAC Address: MAC address of PC1

Result and Analysis

ARP is used to convert an IP address into its corresponding MAC address before communication begins. When the sender does not know the destination MAC address, it broadcasts an ARP Request. The device with the matching IP address responds with an ARP Reply containing its MAC address. The sender stores this information in its ARP cache and then sends the data frame to the correct destination. This process enables successful communication within the local network.

Experiment 4

Aim

To configure two PCs connected through a router, verify connectivity using the **ping** command, and analyze ICMP packets using Wireshark.

Procedure

A simple network was created in Cisco Packet Tracer with two PCs connected through a router. The router interfaces and both PCs were assigned appropriate IP addresses and subnet masks. The default gateways were configured correctly. After completing the configuration, the **ping** command was executed from one PC to the other. Wireshark was used to capture the ICMP packets during communication.

Observation

The ping request was successfully sent, and the destination PC replied with a ping response. The captured packets contained the following ICMP fields:

- **Type:** 8 (Echo Request)
- **Code:** 0
- **Identifier:** Used to identify the ping session
- **Sequence Number:** Increased for every ping request
- **Echo Reply Type:** 0

The **Round Trip Time (RTT)** observed during the experiment was approximately **1 ms**. The RTT is calculated as the time taken for the Echo Request to reach the destination and the Echo Reply to return to the sender.

Result and Analysis

ICMP is mainly used to test network connectivity and troubleshoot communication problems. The Echo Request and Echo Reply messages confirm that the destination device is reachable. The RTT helps measure network performance by showing the time taken for a packet to travel to the destination and back. Since all packets were successfully transmitted and received, the network configuration was correct and communication between the two devices was successful.

Experiment 5

Aim

To create a client-server network, enable HTTP service on the server, access the webpage from the client, and analyze the TCP three-way handshake and HTTP communication using Wireshark.

Procedure

A web server and a client PC were connected in Cisco Packet Tracer. The HTTP service was enabled on the server, and valid IP addresses were assigned to both devices. From the client PC, the web browser was opened and the server's IP address was entered to access the hosted webpage. During this process, Wireshark captured all network packets exchanged between the client and the server.

Observation

Before the webpage was loaded, the TCP three-way handshake was observed.

1. The client sent a **SYN** packet to the server.
2. The server responded with a **SYN-ACK** packet.
3. The client sent an **ACK** packet to complete the connection.

After the connection was established, the following HTTP messages were captured:

- **HTTP GET Request:** Sent by the client to request the webpage.
- **HTTP Response (200 OK):** Sent by the server along with the requested webpage.

The TCP acknowledgements confirmed that all data packets were received successfully.

Result and Analysis

HTTP is an application-layer protocol used to transfer web pages between the client and the server. Since HTTP uses TCP as its transport protocol, the communication

begins with the TCP three-way handshake. TCP ensures reliable communication by establishing the connection, acknowledging received packets, retransmitting lost packets if necessary, and delivering data in the correct order. After the connection is established, the client sends an HTTP GET request, and the server responds with the requested webpage. This experiment demonstrates how the Application Layer and Transport Layer work together to provide reliable web communication.

Conclusion

All the experiments were successfully performed using **Cisco Packet Tracer** and **Wireshark**. The packet captures helped analyze different network protocols such as **Ethernet, IPv4, ARP, ICMP, TCP, and HTTP**. The results showed how different OSI layers work together to provide reliable communication. Packet analysis also helped understand addressing, routing, error checking, and connection establishment, making it easier to visualize how data is transmitted across a network.